

# AHMED BARGOUGUI

AI & Data Engineering | Backend Developer | Computer Science — Final Year  
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## PROFILE

Final-year Computer Science student specializing in AI engineering and backend systems, with two internships at Sagemcom — building a **RAG- and LLM-based Generative AI system for telecom network validation**, and an **ETL/BI pipeline for KPI reporting**. Builder of production-grade AI applications (multi-agent risk analysis, multimodal video search) and backend systems serving **32,000+ users**. Seeking a PFE internship in AI Engineering or Backend Engineering starting February 2027.

## PROFESSIONAL EXPERIENCE

**Telecom Data & AI Engineering Intern** — Sagemcom • Ben Arous, Tunisia 07/2026 – 08/2026

- Built an **ICP/MTP validation** system that extracts data via OCR from telecom compliance documents and generates automated PASS/FAIL reports
- Designed and implemented a Retrieval-Augmented Generation (RAG) pipeline with LLM-based analysis to automatically interpret ICP/MTP validation results, including prompt design and token/context management for reliable, cost-controlled LLM calls
- Layered a **CrewAI** multi-agent system on top of the validation output to generate automated, human-readable interpretation reports
- Engineered an end-to-end **n8n automation** flow — validation → Generative AI analysis → production webhook → automated Gmail reporting — removing manual review from the compliance workflow

**Business Intelligence Engineer Intern** — Sagemcom • Ben Arous, Tunisia 07/2025 – 08/2025

- Cleaned and prepared raw CSV data and validated data requirements ahead of ingestion into the reporting pipeline
- Developed Python scripts for backend-oriented data processing and automation, focusing on the full system flow from raw data to backend processing
- Designed relational database schemas and a **Star Schema** data warehouse; built SQL Server-backed ETL pipelines with Pandas and SQLAlchemy
- Standardised 10,000+ records via Power Query and built **Power BI** dashboards for real-time KPI tracking

## PROJECTS

**AI Risk Analyst — Enterprise Risk Platform** — github.com/Ahmedbrg/Risk\_Analyst 2026

- Built a full-stack enterprise risk platform (FastAPI + React) powered by a **5-agent CrewAI** system that analyzes risk narratives end to end
  - Implemented Retrieval-Augmented Generation with **Qdrant** as the vector store to ground agent outputs in relevant reference material
  - Designed a deterministic, **ISO 31000-aligned scoring engine** so risk scores are reproducible rather than purely LLM-generated
  - Automated the reporting workflow with n8n and added a QC-gated PDF export step to ensure only validated reports are released
- Stack:** Python • FastAPI • CrewAI • Qdrant • RAG • React • n8n • Docker

**ScanVD — AI Video Search Engine** — github.com/Ahmedbrg/ScanVD • huggingface.co/spaces/AhmedBrg/ScanVD\_demo 2026

- Multi-model AI pipeline (**Whisper + YOLOv8 + CLIP**) extracting precise timestamps from natural-language queries over video — deployed live on HuggingFace Spaces
- Stack:** Python • Whisper • YOLOv8 • CLIP • HuggingFace • Docker

## Other Projects

- Forsa:** React Native super-app (community, marketplace, freelance, messaging), JWT-secured Express/MongoDB API.  
github.com/Ahmedbrg/Forsa
- Secure Auth System:** REST API + C# launcher for a 32,000-member Discord community — HWID binding, OAuth2, JWT, rate limiting.

## TECHNICAL SKILLS

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| AI / GenAI: | RAG • LLMs • CrewAI • Qdrant • Whisper • YOLOv8 • CLIP • OCR • Embeddings     |
| Automation: | n8n • Webhooks • Gmail API • Workflow Automation • JSON Pipelines             |
| Backend:    | Node.js • Express • FastAPI • REST APIs • JWT • OAuth2 • MySQL • MongoDB • C# |
| Data / BI:  | SQL Server • ETL • Pandas • SQLAlchemy • Power BI • Star Schema               |
| Languages:  | Python • JavaScript • TypeScript • Java • C++ • C# • SQL                      |
| Tools:      | Git • Docker • Linux • Postman • VS Code • Jupyter                            |

## EDUCATION • LEADERSHIP • LANGUAGES

**Bachelor's Degree in Computer Science** — FSTSBU, University of Kairouan 09/2024 – 06/2027 • Final Year (PFE)

- Club DevUp Founder & President • JCI Sidi Bouzid Trainer • Ranked 3rd ACM Contest • AgriLink (TGN Hackathon 2025)
- Languages:** Arabic (Native) • French (Professional) • English (Professional)